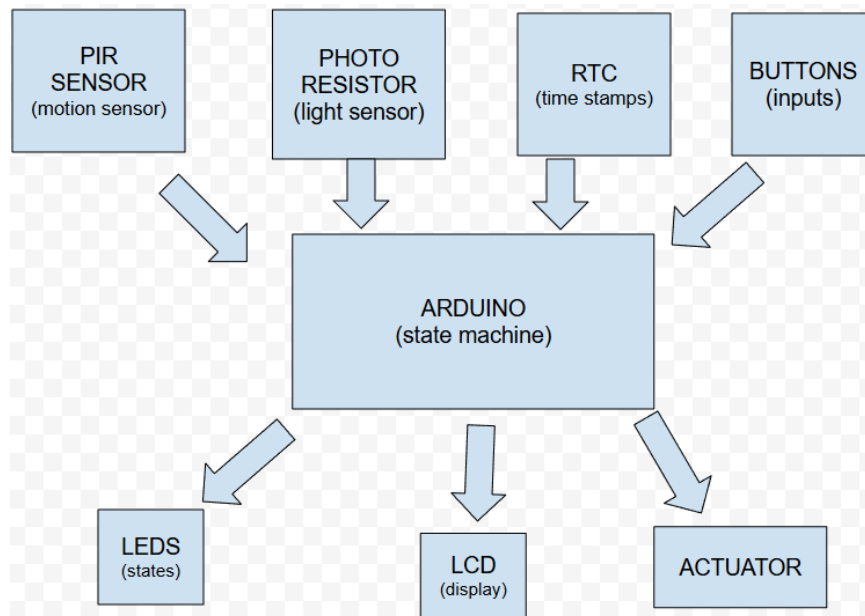


Smart Lighting Control System

Project motivation / Introduction:

Many buildings waste electricity because lights are left on even when nobody is there. This results in wasted energy and electricity costs. This project aims to solve that problem by creating a system that automatically controls lighting based on occupancy and light conditions. The goal of the project is to make a low cost, energy efficient control system to ensure no lights are left on when no one is in the room.

Proposed System:



BLOCK DIAGRAM

Pin mapping:

PIR-D22 , LDR-A0 , RTC-SDA/SCL
LCD-40 to 45 , LEDs-30 to 34
OFF-D23 , ON-D2 , RESET-D24

Component List:

Arduino - used to process all inputs and outputs
PIR Motion sensor - detects motion and uses it as an input
Photoresistor - measures light values and uses it as input
DS1307 RTC - provides real time clock for logging

LCD display - displays system state, light values, and motion values
LEDs(5x) - state indication and actuator simulation
Push Buttons(3x) - ON, OFF, and RESET controls
Potentiometer - sets brightness for LCD display
Resistors - limits current going into components
Breadboard/jumper wires - used to connect and power components

System Behavior:

Modules:

Sensor Module: reads the motion sensor and the photoresistor continuously.

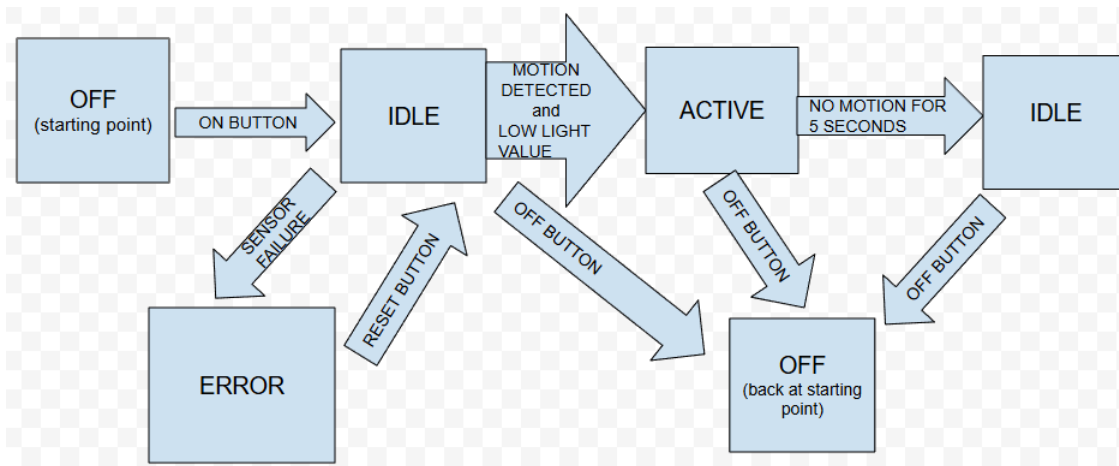
Control Module: handles the transitions from OFF, IDLE, ACTIVE, ERROR. Uses sensor values and button inputs to determine behavior

Display Module: updates the LCD display with the current state, light sensor values, and any error messages. Display changes when system state changes or when light values change

Input Module: Handles the ON, OFF, and RESET inputs.

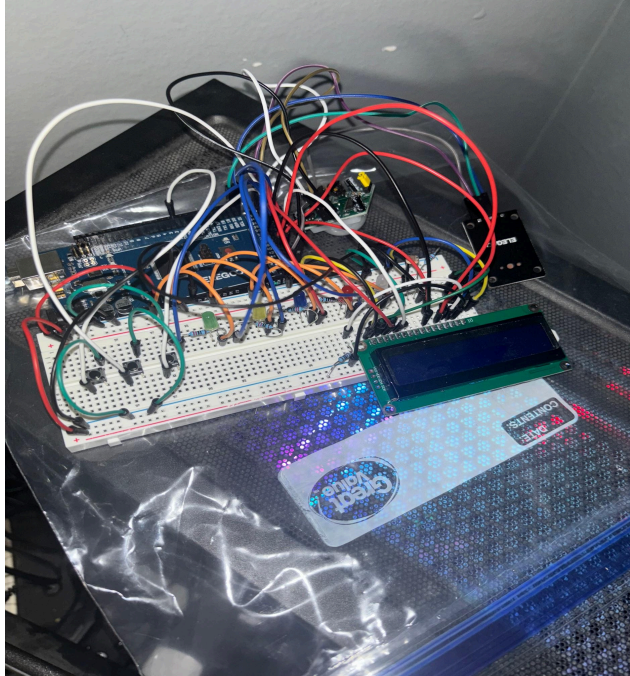
LED Module: Controls the LED for the different states of the system, only one is active at a time. (Green=OFF, Yellow=IDLE, Blue=ACTIVE, Red=ERROR, White=ACTUATOR)

State Diagram:



The system starts in the OFF state, once you press the ON interrupt button, it will go to the IDLE state where it will start monitoring motion and light levels, if motion is detected and light levels are low it will go into the ACTIVE state, once no more motion is detected it will stay in ACTIVE for a few seconds before returning to IDLE. If any sensors are having issues then it will go into an ERROR state, to get back into IDLE you must press the RESET button. You can also press the OFF button to force the OFF state if you want to.

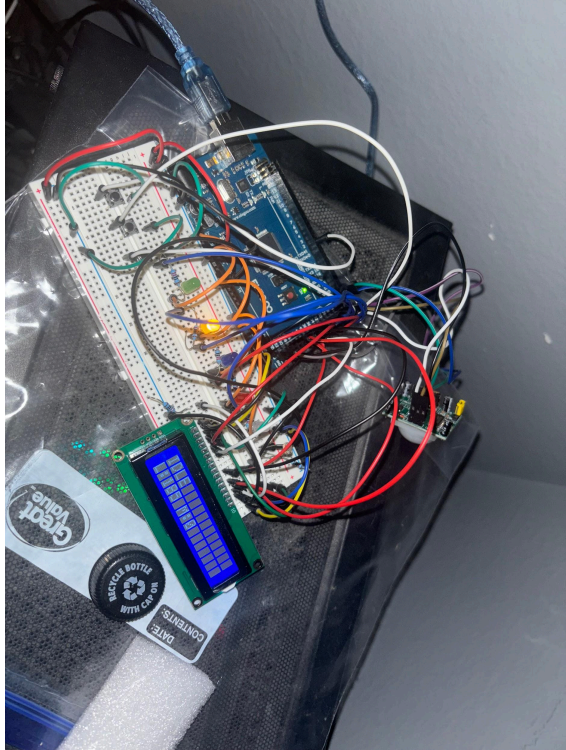
Result:



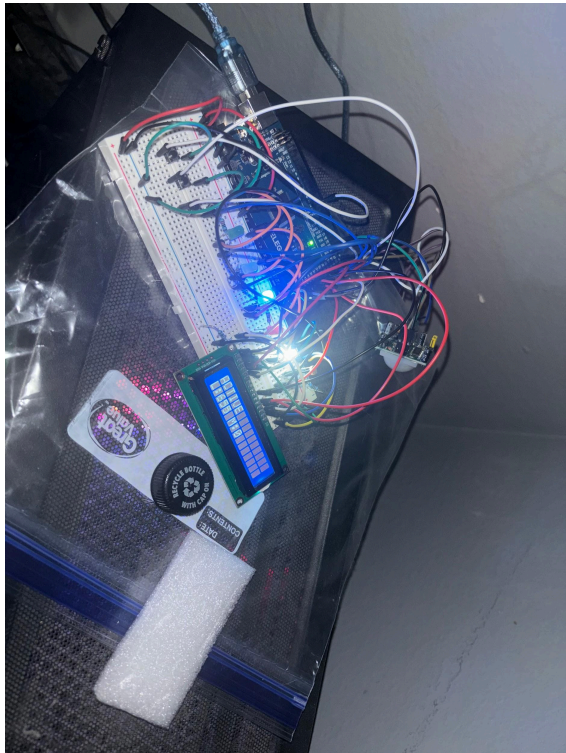
CIRCUIT IMAGE



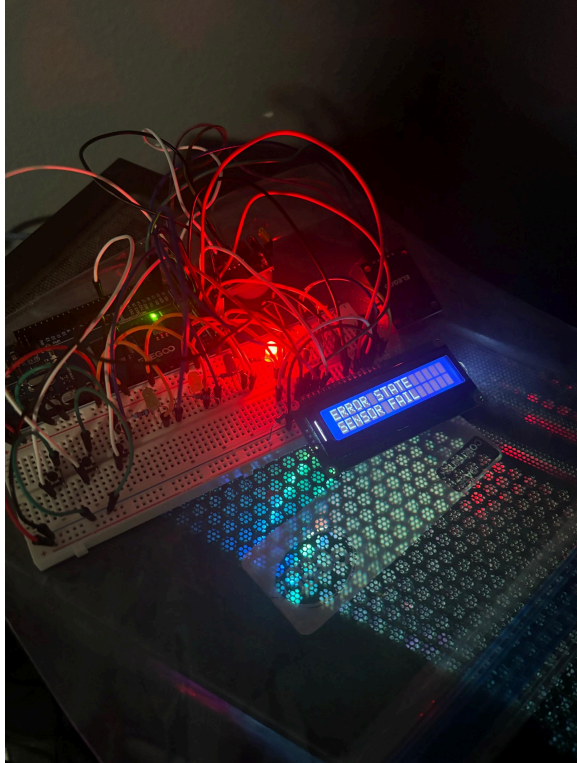
OFF STATE



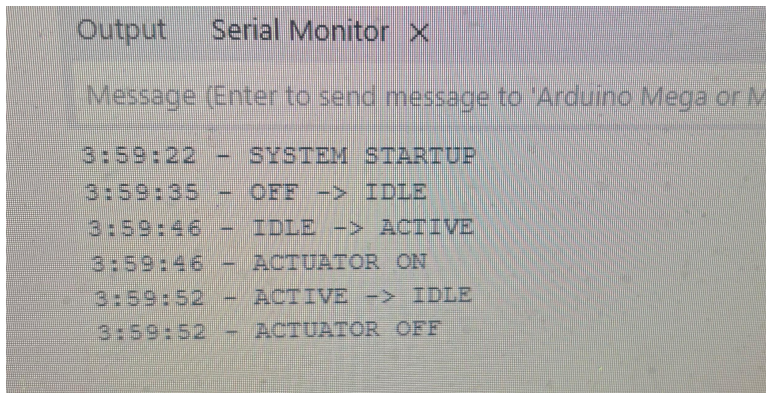
IDLE STATE



ACTIVE STATE



ERROR STATE



SERIAL MONITOR

Environmental & Energy Impact:

Energy Efficiency: The system reduces unnecessary energy usage by turning the lights on only when motion is detected and ambient light values are below a certain level, and also turning off when motion is no longer detected.

Design Safety: The system will operate at a low voltage and it comes with resistors for the LED's, an ERROR state and a RESET function.

Affordability: The system only uses components from our starter kit, no extra purchases are needed.

Sustainability: The system uses a simple design for easy repair and long term use

Accessibility: The system uses a simple design and simple interface with simple to use buttons and LCD messages so that anyone can build and use it, even if they aren't familiar with the technology.

Conclusion:

This project was successful in implementing a smart light controller with real time monitoring. The system uses a state based design with OFF, IDLE, ACTIVE, and ERROR states that are controlled by sensor and button inputs. The sensors used are a motion detector and a light sensor, and the button inputs are ON, OFF, or RESET. The only real limitations of this project is that I wasn't allowed to use certain functions. In the end this project provides a way to reduce wasted electricity by using motion and light detectors to determine if the room light(in this case the white LED) should be on or off, so if nobodies in the room or the room is bright, the light won't be on wasting power